Max Hirsch

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Education

University of California, Berkeley *Aug 2023 - Present*
Ph.D. in Applied Mathematics

- National Science Foundation Graduate Research Fellowship
- Research Interests: Numerical analysis of PDE, multiscale models, inference, model order reduction.
- Advisor: Professor Franziska Weber

Dominican School of Philosophy and Theology *Aug 2025 - Present*
M.A. in Philosophy

- St. Catherine of Siena Scholarship
- Advisor: Fr. Anselm Ramelow, O.P.

Carnegie Mellon University *Aug 2019 – May 2023*
M.S. in Mathematical Sciences, B.S. in Mathematical Sciences

- Thesis : An Energy-Stable Finite Element Scheme for the Q-Tensor Model of Liquid Crystals Subject to an Electric Field
- Advisor: Professor Franziska Weber

École Polytechnique Fédérale de Lausanne *May 2022 - Jan 2023*
Research Internship and Exchange Semester

- Semester Project : Continuous Time Stochastic Gradient Descent for Parameter Identification in Multiscale Diffusions
- Advisor: Professor Fabio Nobile

North Carolina School of Science and Mathematics *Aug 2017 - May 2019*
High School Diploma

Teaching and Mentorship

Graduate Student Instructor, University of California, Berkeley *Spring 2025 - Present*

- Abstract Linear Algebra, Spring 2025

Research Mentor, University of California, Berkeley *Fall 2023 - Present*

- Leon Mikulinsky, Q-Tensor Bayesian Inverse Problem (Summer 2024 - Spring 2025)
- Yijin (Cici) Wang, Physics-Informed Neural Networks for Q-Tensors (Summer 2024)
- Jose Luis Alarcon, Discovering Numerical Schemes from Data (Fall 2023)

Teaching Assistant, Carnegie Mellon University *Fall 2020 - Spring 2023*

- Vector Calculus for Computer Scientists, Spring 2023
- Matrices and Linear Transformations, Fall 2021
- Integration and Approximation, Spring 2021
- Matrices and Linear Transformations, Fall 2020

Publications

*Authorship for most papers is alphabetical.

Preprints

Convergence and Sketching-Based Efficient Computation of Neural Tangent Kernel Weights in Physics-Based Loss *November 2025*
Max Hirsch, Federico Pichi
arXiv preprint arXiv:2511.15530

Nonparametric estimation of homogenized invariant measures from multiscale data via Hermite expansion *October 2025*
Jaroslav Borodavka, Max Hirsch, Sebastian Krumscheid, Andrea Zanoni
arXiv preprint arXiv:2510.25521

Journal Articles

A Convergent Finite Element Scheme for the Q-Tensor Model of Liquid Crystals Subjected to an Electric Field *December 2025*
Max Hirsch, Franziska Weber
Accepted to Journal of Numerical Mathematics arXiv preprint arXiv:2307.11229

Neural empirical interpolation method for nonlinear model reduction *November 2025*
Max Hirsch, Federico Pichi, Jan Hesthaven
SIAM Journal on Scientific Computing 47 (6), C1264-C1293

The zero inertia limit for the Q-tensor model of liquid crystals: analysis and numerics *September 2025*
Max Hirsch, Franziska Weber, Yukun Yue
ESAIM: Mathematical Modelling and Numerical Analysis 59 (5), 2515-2556

Stochastic gradient descent in continuous time for drift identification in multiscale diffusions *November 2024*
Max Hirsch, Andrea Zanoni
ESAIM: Mathematical Modelling and Numerical Analysis 58 (6), 2387-2414

Talks

- Fluid Dynamics Lunch and Learn Seminar at UC Berkeley, *Computing Bifurcation Diagrams for Fluids* (November 2025)
- European Conference on Numerical Mathematics and Advanced Applications (ENUMATH), *Learning the Drift in Multiscale Diffusions* (September 2025)
- mathLab at Scuola Internazionale Superiore di Studi Avanzati (SISSA), *Neural empirical interpolation method for nonlinear model reduction* (July 2025)
- High-Dimensional Scientific Computing Seminar at UC Berkeley, *Neural empirical interpolation method for nonlinear model reduction* (March 2025)
- Computational Math Seminar at UC Berkeley, *Bayesian Inverse Problem for the Q-Tensor Model of Liquid Crystals* (October 2024)
- Applied PDE Student Seminar at UC Berkeley, *Neural empirical interpolation method for nonlinear model reduction* (August 2024)
- European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS), *Neural empirical interpolation method for nonlinear model reduction* (June 2024)
- EPFL Chair of Computational Mathematics and Simulation Sciences Seminar, *Applications of MLniCS : A Library for Physics-Informed Machine Learning for Parametric Problems* (November 2022)
- Finite Element Circus at The Pennsylvania State University, *Finite Elements for the Q-Tensor Flow of Liquid Crystals* (November 2021)

Work Experience

Organist *Berkeley, CA*
Newman Hall-Holy Spirit Parish *February 2025 – Present*

- Play organ and choose the music for two Masses every weekend.

Research Assistant *Remote*
Professor Jay Newby, University of Alberta Department of Mathematical and Sta- *May 2020 – Aug 2021*

tistical Sciences

- Estimated particle densities in videos using convolutional neural networks, achieving 6% error in estimates of the total number of particles from these densities.
- Used Monte Carlo methods for maximum likelihood estimation of the diffusion constant arising from the master equation describing particle motion.

Data Science Intern *Hickory, NC*
Matrix Retail *May 2019 – Aug 2019*

- Created a new scalable sales forecasting algorithm for high-dimensional data which automatically tunes parameters, trains, and forecasts with bootstrapped prediction intervals for over 1000 retail stores to reduce the time of the former forecasting algorithm by 300% without sacrificing accuracy.
- Enabled more efficient evaluation of forecasts by creating a graphical user interface for running the forecasting algorithm and visualizing the resulting forecasts and prediction intervals.

Projects

MLniCS Code Library *May 2022 - January 2022*

- Developed MLniCS, a code library for building physics-informed neural networks for reduced order parameterized partial differential equations.
- MLniCS Repository

Tetris Reinforcement Learning (Project Leader) *Spring 2020 - Fall 2020*

- Led a Data Science Club project group in which we trained a reinforcement learning model with the REINFORCE algorithm and Monte Carlo tree search to play Tetris.

Duke Datafest Rugby Player Health Prediction *April 2019*

- Implemented a neural network, random forest, and support vector machine to determine the effect of a workout on Canadian women's rugby players health with over 70% accuracy.

MathWorks Math Modeling Challenge (1st Place, Technical Computing) *March 2019*

- Classified likely high school student drug users based on demographic data with average accuracy of 77% across all drugs in the study. Implemented the Metropolis-Hastings algorithm as part of a model to predict the spread of drug use in high school students.

Reducing Exam Conflicts with Hill Climbing (Implemented Fall 2019) *February 2019*

- Designed a hill climbing algorithm which reduced the number of exam conflicts at the North Carolina School of Science and Mathematics to half the number of conflicts while maintaining schedule structure. Implemented for the 2019 fall trimester.

Honors and Awards

- April 2025 - St. Catherine of Siena Scholarship, Dominican School of Philosophy and Theology
- May 2024 - Conference Travel Grant, UC Berkeley Graduate Division
- April 2023 - Mathematics Prize for outstanding scholastic achievement during undergraduate career, CMU Dept. of Mathematical Sciences
- April 2023 - Nominated to Phi Beta Kappa Society
- March 2023 - National Science Foundation Graduate Research Fellowship
- May 2022 - Richard A. Moore Award for outstanding scholastic achievement, CMU Dept. of Mathematical Sciences
- April 2022 - Grant to attend HYP2022 Conference on Hyperbolic Problems
- March 2022 - ThinkSwiss Research Scholarship
- February 2022 - EPFL Research Internship Scholarship of Excellence
- May 2021 - Richard A. Moore Award for outstanding scholastic achievement, CMU Dept. of Mathematical Sciences
- November 2019 - CMU Data Science Club November Challenge Winner
- May 2019 - CMU Quantitative Social Science Scholar (QSSS)

- March 2019 - MathWorks Math Modeling Challenge (M3C) Honorable Mention, 1st Place in Technical Computing
- February 2019 - Mathematical Contest in Modeling (MCM) Honorable Mention

Skills

Languages: English (Native), French (Proficient)

Computer Languages: **Python**, C, MATLAB, SML, Java, R

Libraries: Pandas, **Numpy**, Scikit Learn, Matplotlib, Numba, **PyTorch**, TensorFlow

Tools: Git, Vim, LaTeX

Organizations

- Thomistic Institute at UC Berkeley, President (2025-present); Society of Catholic Scientists, Member (2023-present); Carnegie Mellon Catholic Newman Club, President (2021-2022); Carnegie Mellon Data Science Club, Vice President of Projects (2020-2021)

Outreach

- Society of Catholic Scientists St. Albert Initiative, *Big Questions/Ask a Scientist (Invited Poster)* (March 2026)
- Dominican School of Philosophy and Theology, College of Fellows Convocation Paper Presentation, *Education and Artificial Intelligence* (January 2026)
- Dominican School of Philosophy and Theology, Thomistic Lyceum, *What is philosophy, and do we do it?* (November 2025)